



Data-Driven Decision Support for Aircraft Procurement

Student: Daniel Mwaka

Student Pace: DSF-FT12-Hybrid

Instructor: Samuel Karu

Overview

This project aims to support the company's data-driven decision to procure relatively low-risk airplane models to operate the fleet in the commercial and private enterprise sectors. It analyzes a dataset compiled by the National Transportation Safety board and retrieved from Kaggle.com. Presented recommendations are deduced from descriptive statistics results and backed-up by interactive visualizations modeled using Tableau Public.

Business Problem

Venturing into a new industry avails a company of potential growth opportunities. Diversifying a company's portfolio lowers reliance on a single revenue stream (asset diversification) and optimizes resilience against unfavorable business environments/ factors (Luo, 2022). However, venturing into a highly sensitive sector, such as operating airplanes for commercial and private enterprises, necessitates data-driven decisions, strategic implementation, and formative performance evaluation (Altundag & Wynn, 2024). Purchasing airplane models prone to crashing poses a substantial risk to human life, brand image, competitiveness, profitability, and the company's longevity once it ventures into the new industry.

Data Understanding

The NTSB aviation accident database contains information on crashes and contingency incidents within the U.S., its territories, and across international waters from 1962 to 2023.

The dataset was retrieved from Kaggle.com (NTSB, 2023). The dataset is formatted as a .csv file with 31 columns and 88889 rows. The columns are meticulously organized to capture variables of interest in aircraft accidents and incidents. Extracting, analyzing, and visualizing relevant data from the dataset is vital to shedding insight into the safest, low-risk airplane models for a new entrant to the commercial aviation industry.

However, the dataset is enormous, contains some missing values duplicates, and needs to be customized to meet new entrants' needs aiming to purchase and operate airplanes for commercial and private enterprises. For instance, it contains data for airplane crashes across multiple scenarios (purpose of the flight), in which some cases exceed the target scope of the company. For this project, the target variables are categorized into Dimensions and Measures. The Dimension variables include: Event Date, Investigation Type, Aircraft Damage, Location, Make, Model, Weather Condition, and Purpose of Flight. The Measure variables include: The number of Engines, Total Fatal Injuries, Total Minor Injuries, and Total Uninjured passengers. The insights yielded from analyzing these variables and visualizing their respective relationships are deemed appropriate for deducing recommendations to support data-driven decisions by the company to procure a fleet that comprises safe, low-risk airplanes

Data Preparation

Digital maturity in leveraging novel data analytics is the key driver for aircraft safety. Manufacturers use these technologies to design safer aircraft (Boyd & Stolzer, 2016). On the other hand, companies operating in the airline sector base their strategic procurement plans on data-supported decisions (Altundag & Wynn, 2024). The progressive scope in which key stakeholders in the airline sector are embracing data analytics is reflected in the cumulative number of aircraft accidents and incidents recorded by the NTSB for each decade. Additionally, structural and material design advancements progressively increase aircraft safety.

As captured in the time-series plot, the number of aircraft accidents and incidents has been dropping. The selected period is deemed appropriate because analyzing data for aircraft accidents and incidents before 2000 would comprise the reliability of deduced recommendations for real-world application in the 2020s. The next step is to drop all the columns deemed inappropriate for this project

Data Modeling

```
In [10]: import pandas as pd
from src.data_loader import load_data, initial_inspection
from src.data_cleaning import clean_data
from src.data_analysis import plot_safest_aircraft, plot_riskiest_ai

# Load and inspect data
df_master = load_data("data/aviation-data.csv")
df = df_master.copy()
```

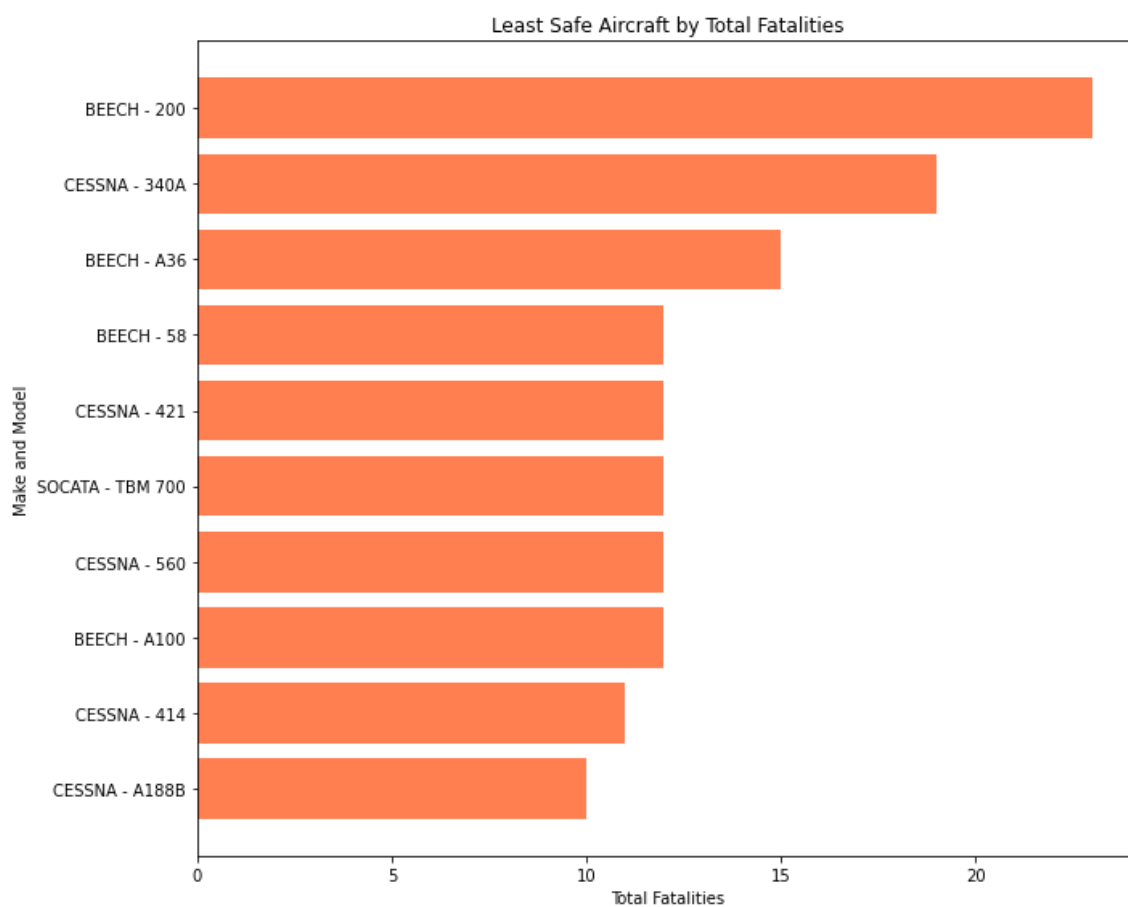
```
In [12]: # Inspecting the data
df.info()
```

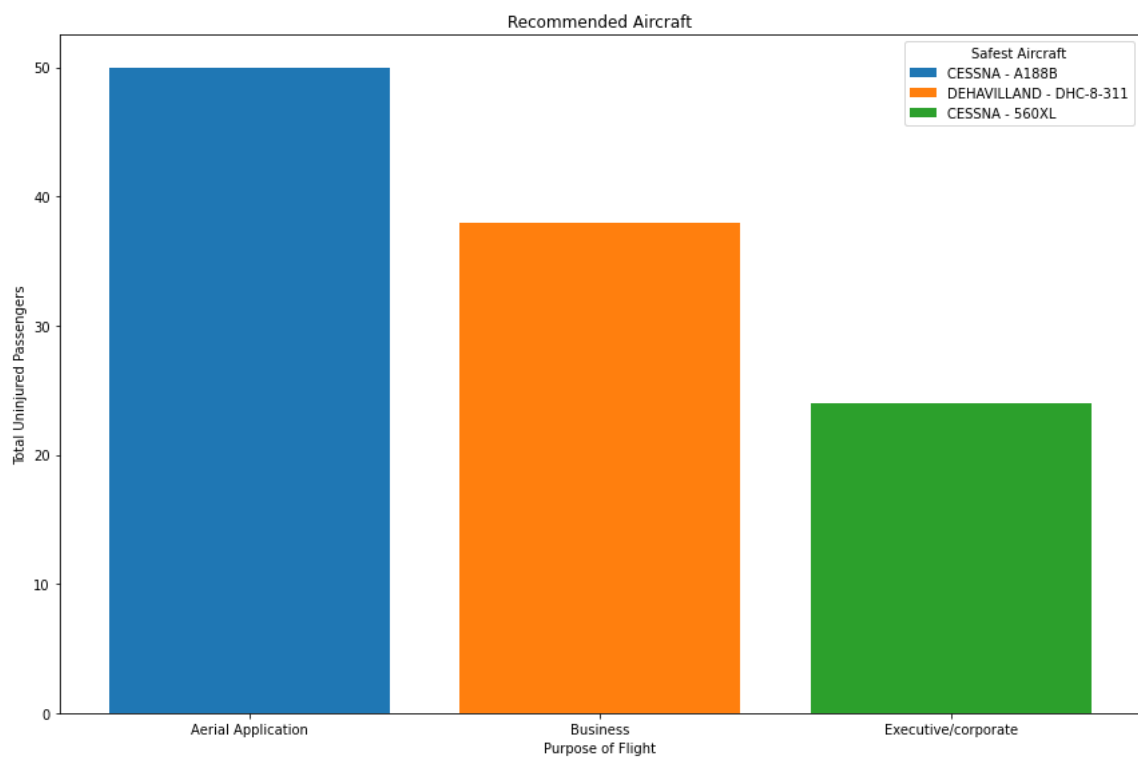
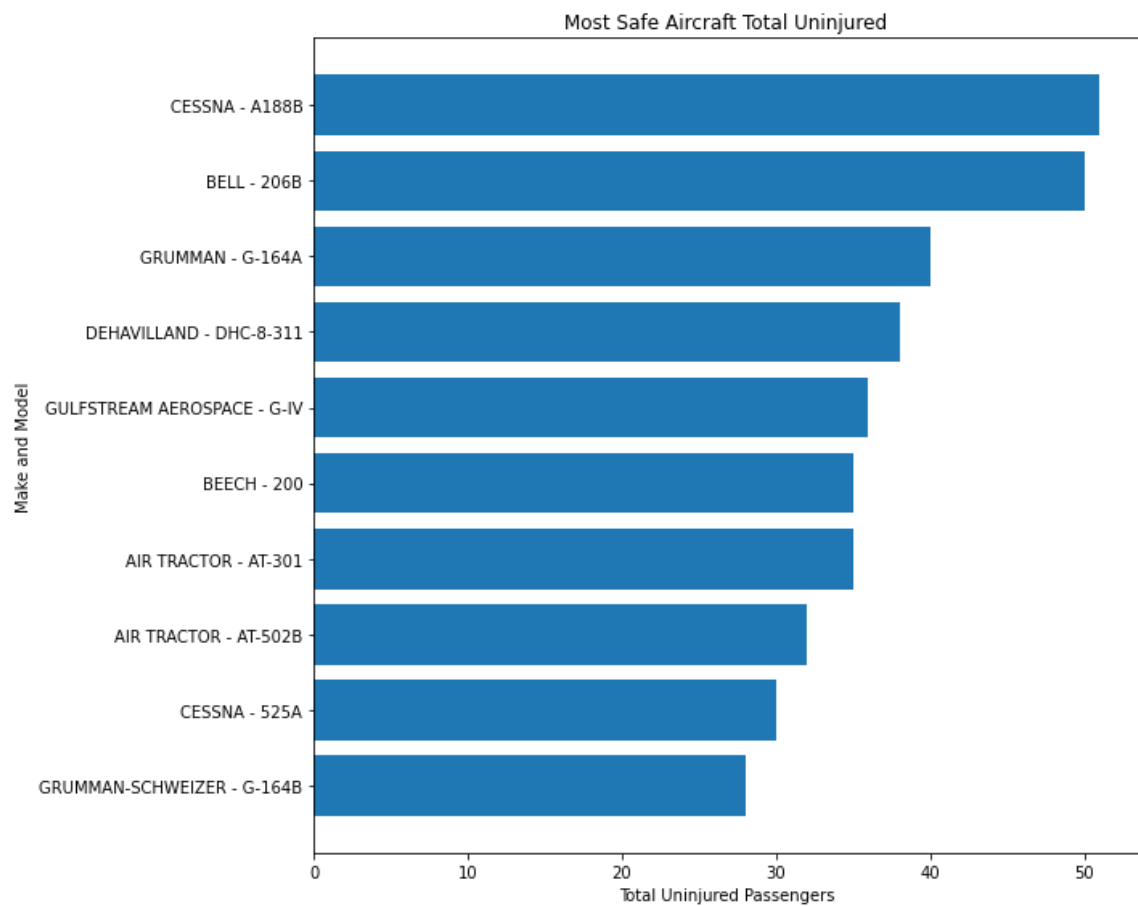
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 88889 entries, 0 to 88888
Data columns (total 31 columns):
 #   Column                                  Non-Null Count  Dtype
---  -
 0   Event.Id                               88889 non-null  object
 1   Investigation.Type                     88889 non-null  object
 2   Accident.Number                        88889 non-null  object
 3   Event.Date                             88889 non-null  object
 4   Location                               88837 non-null  object
 5   Country                               88663 non-null  object
 6   Latitude                               34382 non-null  object
 7   Longitude                              34373 non-null  object
 8   Airport.Code                           50249 non-null  object
 9   Airport.Name                           52790 non-null  object
10   Injury.Severity                        87889 non-null  object
11   Aircraft.damage                        85695 non-null  object
12   Aircraft.Category                      32287 non-null  object
13   Registration.Number                    87572 non-null  object
14   Make                                   88826 non-null  object
15   Model                                  88797 non-null  object
16   Amateur.Built                          88787 non-null  object
17   Number.of.Engines                      82805 non-null  float64
18   Engine.Type                            81812 non-null  object
19   FAR.Description                        32023 non-null  object
20   Schedule                               12582 non-null  object
21   Purpose.of.flight                      82697 non-null  object
22   Air.carrier                             16648 non-null  object
23   Total.Fatal.Injuries                   77488 non-null  float64
24   Total.Serious.Injuries                 76379 non-null  float64
25   Total.Minor.Injuries                   76956 non-null  float64
26   Total.Uninjured                        82977 non-null  float64
27   Weather.Condition                      84397 non-null  object
28   Broad.phase.of.flight                  61724 non-null  object
29   Report.Status                          82508 non-null  object
30   Publication.Date                       75118 non-null  object
dtypes: float64(5), object(26)
memory usage: 21.0+ MB
```

```
In [13]: # Clean the data
df = clean_data(df)
df.to_csv("data/cleaned-aviation-data.csv", index=False, encoding='latin1')
df_clean = pd.read_csv("data/cleaned-aviation-data.csv", encoding='latin1')
```

```
In [15]: # Analyze and plot data for all aircrafts regardless of number of eng
print("Plotting for all aircrafts...")
plot_riskiest_aircraft(df_clean)
plot_safest_aircraft(df_clean)
plot_recommended_aircraft(df_clean)
```

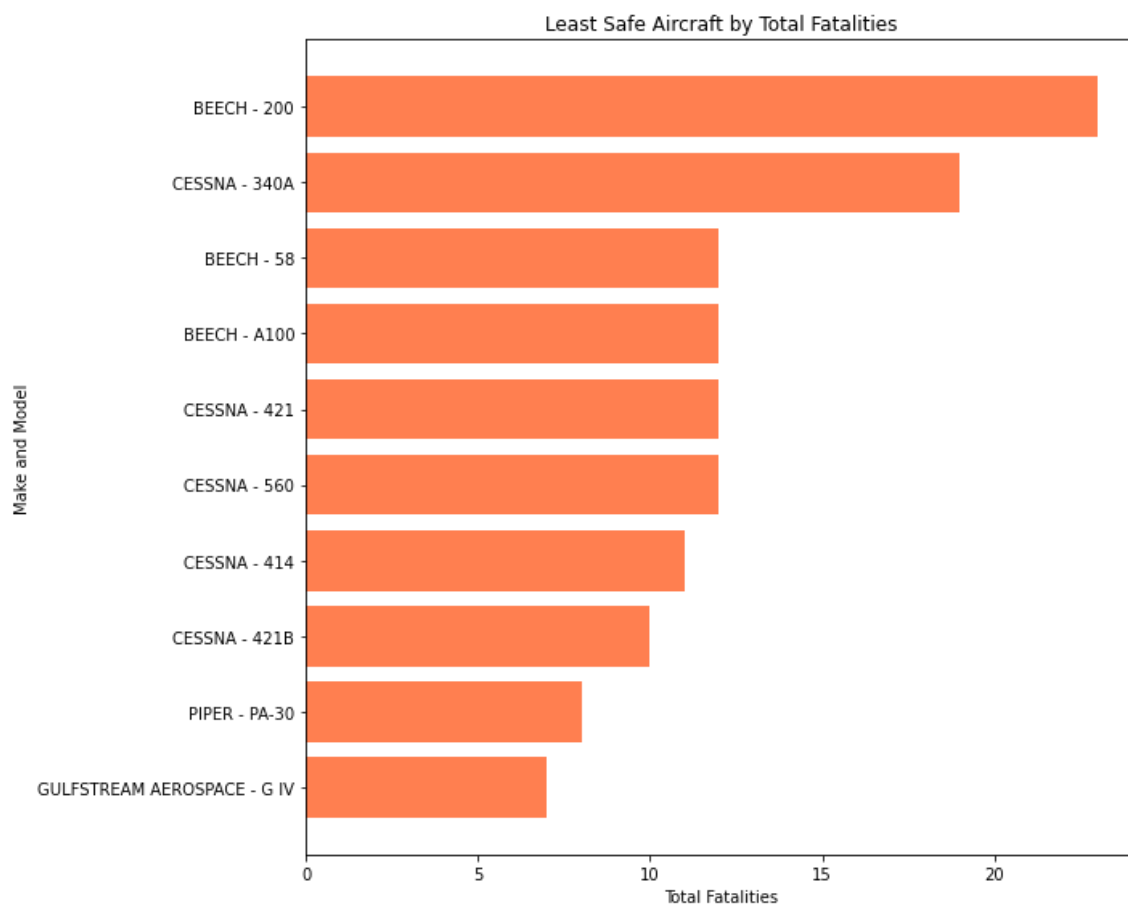
Plotting for all aircrafts...

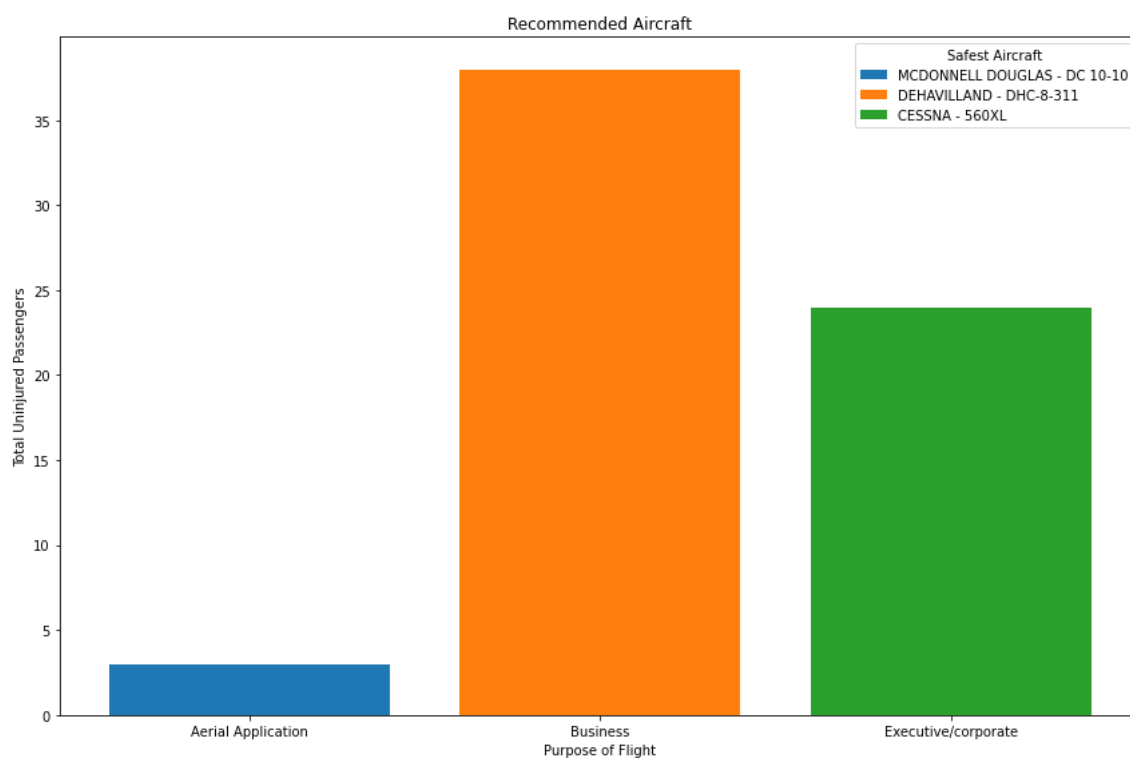
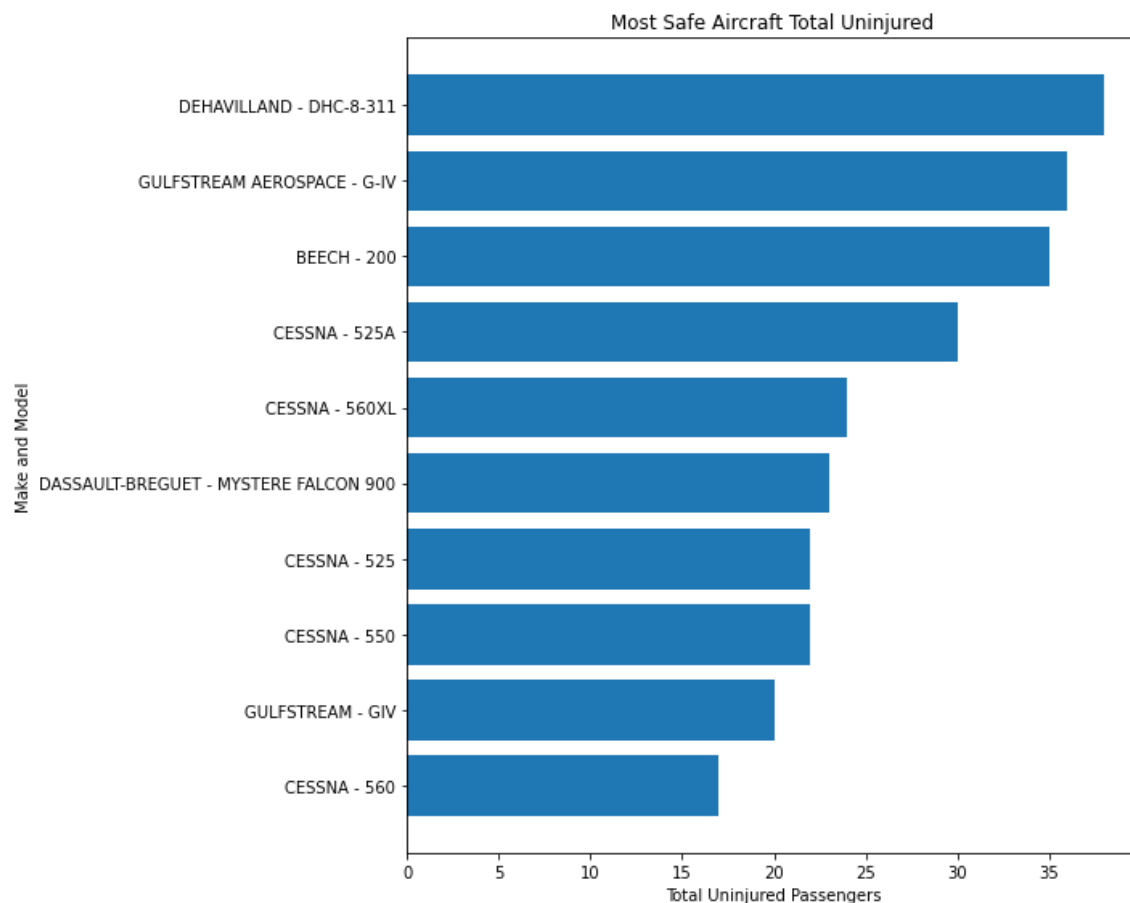




Multi-engine aircraft are typically safer in comparison to single-engine airplanes (Pilot Institute, 2023). More than one engine avails redundancy to propulsion units. Pilots can recalibrate controls and continue flying if one engine stalls or malfunctions. In contrast, such an incident in a single-engine aircraft poses a substantial risk of crashing if the gliding distance falls short of a runway or a relatively flat- surface for an emergency landing. Thus, I modified the baseline model to drop row entries whose `Number.of.Engines` is less than 2.

```
In [16]: # Analyze and plot data for multi-engine aircraft
df_modified = df_clean.copy()
df_modified = df_modified[df_modified['Number.of.Engines'].apply(lambda x: x > 1)]
plot_riskiest_aircraft(df_modified, multi_engine=True)
plot_safest_aircraft(df_modified, multi_engine=True)
plot_recommended_aircraft(df_modified, multi_engine=True)
```





Evaluation

The baseline model shed the following insights:

- The BEECH-200 (8-seater), the CESSNA-340A (6-seater), and the BEECH-A36 (6-seater), are the top-three most risky aircraft overall.
- The CESSNA-A188B (1-seater), the BELL-206B (5-seater), and the GRUMMAN-

G-164A (1-seater) are the top-three safest aircraft models overall.

Recommendations:

- The CESSNA-560XL (10-seater) is the safest aircraft for Executive/corporate flights.
- The DEHAVILLAND-DHC-8-311 (50-seater) is the safest airplane for business flights.
- The CESSNA-A188B (1-seater) is the safest aircraft for Aerial Applications.

The number of engines for these aircraft models.

- BEECH-200: 2 engines
- CESSNA-340A: 2 engines
- BEECH-A36: 1 engine
- BELL-206B: 1 engine
- GRUMMAN-G-164A: 1 engine
- CESSNA-560XL: 2 engines
- CESSNA-A188B: 1 engine
- DEHAVILLAND-DHC-8-311: 2 engines

The modified model sheds the following insight:

- The BEECH-200 (13-seater), the CESSNA-340A (6-seater), and the BEECH-A400 (8-seater) are the top-three most risky multi-engine aircraft to operate.
- The DEHAVILLAND DHC-8-311 (50-seater), the GULFSTREAM AEROSPACE-G-IV (10-seater), and the BEECH-200 (13-seater) are the top-three safest multi-engine aircraft to operate.

Recommendations:

- The CESSNA 560XL(10-seater) is the safest multi-engine aircraft for Executive/corporate flights.
- The DEHAVILLAND -DHC-8-311(50-seater) is the safest multi-engine aircraft for business flights.
- The MCDONNELL DOUGLAS-DC 10-10(250-seater) is the safest multi-engine aircraft for Aerial Applications.

The number of engines for these aircraft models.

- BEECH-200: 2 engines
- CESSNA-340A: 2 engines
- BEECH-A400: 2 engines
- GULFSTREAM AEROSPACE-G-IV: 2 engines
- CESSNA 560XL: 2 engines
- DEHAVILLAND DHC-8-311: 2 engines
- MCDONNELL DOUGLAS -DC 10-10: 3 engines

The findings from the results yielded by the baseline model are compared to those yielded by the modified model. After entering the industry, the recommended safest aircraft model for the aviation services the company will offer is based on respective applicability and operational logistics.

Conclusions

The analysis yields three recommendations on the aircraft models the company should procure and operate after entering the commercial aviation industry.

- **The CESSNA-560XL aircraft is recommended for Executive/ Corporate flights:** The baseline and modified model conform the aircraft is safest for executive and corporate flights.
- **The DEHAVILLAND DHC-8-311 aircraft is recommended for Business Flights:** The baseline and modified model conform the aircraft is safest for business flights.
- **The CESSNA-A188B aircraft is recommended for Aerial Applications:** The modified model proposes the MCDONNELL DOUGLAS-DC 10-10 (a three-engine, 250-seater) aircraft for aerial applications. The proposed alternative by the modified model is rejected because aerial applications typically include agricultural activities such as spraying crop fields. Hence, the single-engine CESSNA-A188B is recommended for aerial applications.

References

- Altundag, A., & Wynn, M. (2024). Advanced Analytics and Data Management in the Procurement Function: An Aviation Industry Case Study. *Electronics*, 13(8), 1554. <https://doi.org/10.3390/electronics13081554> (<https://doi.org/10.3390/electronics13081554>)
- Boyd, D. D., & Stolzer, A. (2016). Accident-precipitating factors for crashes in turbine-powered general aviation aircraft. *Accident Analysis & Prevention*, 86, 209-216. <https://doi.org/10.1016/j.aap.2015.10.024> (<https://doi.org/10.1016/j.aap.2015.10.024>)
- NTSB. (2023). Aviation Accident Database & Synopses, up to 2023. <https://doi.org/10.34740/KAGGLE/DSV/4875576> (<https://doi.org/10.34740/KAGGLE/DSV/4875576>)
- Luo, J. (2022). Data-driven innovation: What is it?. *IEEE Transactions on Engineering Management*, 70(2), 784-790. <https://doi.org/10.1109/TEM.2022.3145231> (<https://doi.org/10.1109/TEM.2022.3145231>)
- Pilot Institute. (2023). Single Engine Vs. Multi Engine: Which is Better? <https://pilotinstitute.com/single-vs-multi-engine/> (<https://pilotinstitute.com/single-vs-multi-engine/>)